

The nuclear receptor Ftz-F1 and homeodomain protein Ftz interact through evolutionarily conserved protein domains

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Received 11 May 2001; accepted 29 May 2001

Abstract

The *Drosophila* homeodomain protein Fushi Tarazu (Ftz) and its partner, the orphan receptor Ftz-F1, are members of two distinct families of DNA binding transcriptional regulators. Ftz and Ftz-F1 form a novel partnership in vivo as a Hox/orphan receptor heterodimer. Here we show that the murine Ftz-F1 ortholog SF-1 functionally substitutes for Ftz-F1 in vivo, rescuing the defects of *ftz-f1* mutants. This finding identified evolutionarily conserved domains of Ftz-F1 as critical for activity of this receptor in vivo. These domains function, at least in part, by mediating direct protein interactions with Ftz. The Ftz-F1 DNA binding domain interacts strongly with Ftz and dramatically facilitates the binding of Ftz to target DNA. This interaction is augmented by a second interaction between the AF-2 domain of Ftz-F1 and the N-terminus of Ftz via an LRALL sequence in Ftz that is reminiscent of LXXLL motifs in nuclear receptor coactivators. We propose that Ftz-F1 serves as a cofactor for Ftz by facilitating the selection of target sites in the genome that contain Ftz/Ftz-F1 composite binding sites. Ftz, on the other hand, influences Ftz-F1 activity by interacting with its AF-2 domain in a manner that mimics a nuclear receptor coactivator. © 2001 Elsevier Science Ireland Ltd. All rights reserved.

Keywords: Transcription regulation; Nuclear hormone receptor; Homeodomain; Ftz; SF-1

1. Introduction

Hox genes determine cell fate and specify regional identity during animal development. These genes are found in evolutionarily conserved clusters throughout the animal kingdom (reviewed in McGinnis and Krumlauf, 1992). In *Drosophila*, *Hox* genes are clustered in the Antennapedia and Bithorax complexes, together comprising HOM-C. *Hox* genes encode transcriptional regulatory proteins that act as molecular switches to control the expression of sets of downstream target genes, most of which have not yet been identified. Each *Hox* gene has a unique and specific function in vivo. However, all *Hox* proteins contain an ‘Antp-class’ homeodomain (HD) and isolated *Hox* proteins bind to similar DNA sequences in vitro (Florence et al., 1991; Gehring et al., 1994; McGinnis et al., 1984; Scott and Weiner, 1984). The DNA target recognition sequence is small and flexible, such that *Hox* proteins have the potential to interact with an

unusually large number of sites in the genome. This virtually ubiquitous DNA binding is at odds with the exquisitely circumscribed biological specificity of *Hox* proteins. One explanation for this paradox is that HD proteins interact with specific cofactors to increase binding site selectivity or transcriptional activity, thereby conferring functional specificity in vivo (reviewed in Biggin and McGinnis, 1997; Mann, 1995).

The *Drosophila homeobox* gene *fushi tarazu* (*ftz*) is located in HOM-C and Ftz protein has a canonical Antp-class HD (Kuroiwa et al., 1984; Weiner et al., 1984). Unlike the homeotic functions typical of most HOM-C members, *ftz* is a pair-rule segmentation gene, necessary for the establishment of even numbered parasegments (Nusslein-Volhard et al., 1984; Wakimoto and Kaufman, 1981). As such, Ftz should regulate a very different set of downstream target genes from other HOM-C proteins. To determine how the unique functions of Ftz are achieved in vivo, we set out several years ago to identify specific partner(s) of Ftz that modulate its DNA binding and/or transcriptional abilities in the embryo (Yu et al., 1997, 1999). These studies identified Ftz-F1, an orphan nuclear hormone receptor family member, as a novel partner of Ftz protein. *ftz-f1* mutant embryos display *ftz*-like pair-rule phenotypes and lack Ftz-dependent *engrailed* stripes (Florence et al., 1997;

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Guichet et al., 1997; Yu et al., 1997). This demonstrated that Ftz-F1 is absolutely required as a cofactor for Ftz function in vivo. Composite DNA binding sites for Ftz-F1 and Ftz were identified in two bona fide Ftz target regulatory elements: the *ftz* upstream autoregulatory element (Yu et al., 1997) and a Ftz-dependent *engrailed* regulatory element (Florence et al., 1997). Ftz and Ftz-F1 bind these elements cooperatively, with Ftz-F1 dramatically enhancing specific DNA binding by Ftz protein. This indicated that Ftz-F1 influences Ftz function primarily at the level of DNA target site selection in vivo and further suggested that Ftz and Ftz-F1 would coordinately activate transcription of specific target genes via composite binding sites in vivo.

Ftz is expressed in a tightly controlled seven stripe pattern in the embryo that correlates with the *ftz* mutant phenotype (Hafen et al., 1984; Yu and Pick, 1995). In contrast, Ftz-F1 is found in the nuclei of all somatic cells of the blastoderm embryo (Yu et al., 1997 and see Fig. 6). The *ftz*-like pair-rule phenotype displayed by *ftz-fl* mutant embryos was rather surprising in light of its ubiquitous expression pattern, raising the possibility that localized Ftz expression in stripes is the limiting factor enabling activation of transcription by Ftz-F1. How is this activation of Ftz-F1 achieved? Ftz-F1 is an orphan nuclear receptor with orthologs found in a wide variety of animals including *Drosophila* (Lavorgna et al., 1991; Ueda et al., 1990), *Bombyx mori* (Ueda and Hirose, 1991), *C. elegans* (Gissendanner and Sluder, 2000), zebrafish (Chai and Chan, 2000; Liu et al., 2000), *Xenopus* (Ellinger-Ziegelbauer et al., 1994), mouse (Lala et al., 1992, reviewed in Hanley et al., 2000) and human (Taketo et al., 1995). Ftz-F1 proteins appear to be constitutively nuclear and bind DNA as monomers with a relatively stringent consensus sequence (Ellinger-Ziegelbauer et al., 1995; Han et al., 1993; Lala et al., 1992; Ueda et al., 1992). In *Drosophila*, two alternatively spliced forms of Ftz-F1 (Lavorgna et al., 1993) are expressed at different developmental stages: α Ftz-F1 (studied here) is maternally deposited and required for segmentation. β Ftz-F1 is required in the larva as a competence factor for metamorphosis (Broadus et al., 1999). Ftz-F1 proteins have structures typical of nuclear receptors: a variable N-terminal A/B region, followed by a conserved zinc finger DNA binding domain (DBD) region C, a hinge region D, and ligand binding domain (LBD, region E) (reviewed in Mangelsdorf et al. (1995) and Fig. 2). The C-terminus of the nuclear receptor LBD contains the activation function 2 domain (AF-2) that is critical for ligand-dependent transcription activation: ligand binding activates nuclear receptors by causing a dramatic conformational change in the AF-2 domain, allowing it to interact with repeated LXXLL motifs in coactivators such as SRC or CBP/p300 (Baretino et al., 1994; Danielian et al., 1992; Darimont et al., 1998; Durand et al., 1994; Heery et al., 1997; McNerney et al., 1998; Nolte et al., 1998; Tone et al., 1994). For orphan receptors such as Ftz-F1, ligands have not been identified and the mechanism of activation is less clear. This mechanism can

be studied in *Drosophila* where the simple 'on' versus 'off' state of Ftz-F1 in Ftz expressing versus Ftz-non-expressing cells in the blastoderm provides a system to dissect the requirements for Ftz-F1 function in vivo.

Here we show that evolutionarily conserved regions of Ftz-F1 are required for protein function in vivo and for physical interaction with Ftz protein in vitro. Ftz and Ftz-F1 form stable complexes on DNA and coordinately activate transcription. Murine SF-1, an ortholog of the late form of *Drosophila* Ftz-F1, substitutes for *Drosophila* Ftz-F1 in transgenic animals, rescuing the embryonic defects of *ftz-fl* mutants, and synergistically activates transcription with Ftz in vitro. The highly conserved DBD of Ftz-F1 interacts strongly with Ftz protein. In addition, Ftz-F1 utilizes its AF-2 domain to interact with Ftz via a coactivator-type LXXLL motif that Ftz has acquired. These findings suggest a novel mechanism for Ftz-F1 activation via interaction with a Hox protein that has usurped one function of a nuclear receptor coactivator. A model for how Ftz and Ftz-F1 interact to coordinately regulate gene expression in vivo is presented.

2. Results

2.1. Murine SF-1 rescues the α ftz-fl mutant phenotype

As described above, loss of maternal α ftz-fl results in defects indistinguishable from those seen for mutations in *ftz* itself (Yu et al., 1997; Guichet et al., 1997; Florence et al., 1997). These embryos fail to hatch and exhibit *ftz*-like pair-rule defects (see Fig. 1A). In addition, *ftz-fl* mutant embryos express only seven *engrailed* stripes rather than the 14 stripes seen in wild type embryos, due to loss of the seven Ftz-dependent stripes (Fig. 1B). To test the ability of various natural isoforms of Ftz-F1 to rescue these phenotypes, we used the binary UAS/GAL4 system (Brand and Perrimon, 1993) with the NGT40 GAL4 driver that directs ubiquitous gene expression in blastoderm embryos (Tracey et al., 2000). This ubiquitous expression in blastoderm stage embryos is similar to the expression pattern of endogenous α ftz-fl. Accordingly, expression with this GAL4 driver in a wild type background did not lead to phenotypic alterations (not shown). Females homozygous for the maternal-specific allele *ftz-fl*²⁰⁹ (Guichet et al., 1997) carrying the NGT40 GAL4 driver were crossed with different UAS lines. The expression of a UAS-lacZ transgene had no effect on *ftz-fl* mutant phenotypes (Fig. 1A,B). However, expression of α ftz-fl with the NGT40 GAL4 driver fully rescued the cuticle defects (Fig. 1C) associated with α ftz-fl mutant embryos and restored expression of Engrailed in 14 stripes (Fig. 1D).

The late, zygotically expressed form of Ftz-F1, β Ftz-F1, is a variant that differs from α Ftz-F1 only in the A/B region (see Fig. 2A). Nevertheless, as shown in Fig. 1E,F, expression of β ftz-fl also fully rescued the maternal *ftz-fl* pheno-

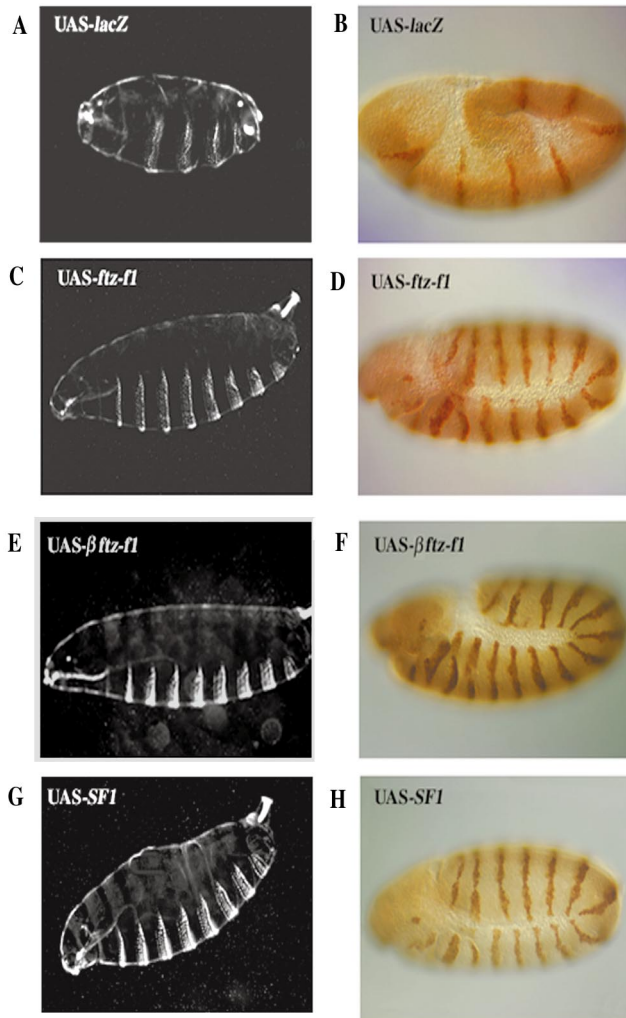


Fig. 1. $\beta ftz-f1$ and mouse *SF-1* can rescue the $\alpha ftz-f1$ embryonic phenotype. The UAS-GAL4 system was used to test the ability of $\beta ftz-f1$ and *SF-1* to rescue embryonic *ftz-f1* mutant phenotypes. *ftz-f1*²⁰⁹ mutant females carrying the NGT40 Gal4 driver were crossed to different UAS lines, as indicated. The embryos were analyzed for rescue of cuticle defects (A,C,E,G) and expression of the Ftz-F1/Ftz target gene *engrailed* (B,D,F,H). The expression of the *lacZ* gene had no effect on the *ftz-f1* mutant phenotype (A,B). The expression of $\alpha ftz-f1$ fully restored a wild type cuticle pattern (C) and expression of *Engrailed* in 14 stripes (D). Both $\beta ftz-f1$ and *SF-1* restored wild type expression of *Engrailed* in 14 stripes (F,H) and a normal cuticle pattern (E,G).

type. Even more striking was the finding that the mouse ortholog of late β Ftz-F1, *SF-1* (see Fig. 2A) also rescued both the *Engrailed* expression pattern (1H) and the cuticular phenotype associated with $\alpha ftz-f1$ mutant embryos (1G). Thus, the molecular function of Ftz-F1 proteins has been conserved in animals as divergent as fly and mouse. These results demonstrate that evolutionary conserved regions of Ftz-F1 are critical for function of the *Drosophila* protein in vivo.

The Ftz-F1 proteins tested here share two highly conserved regions: the DBD and putative LBD (Fig. 2A). The zinc finger DBD is highly conserved throughout the

nuclear receptor superfamily (Mangelsdorf et al., 1995). A 'Ftz-F1' box at the C-terminus of the DBD distinguishes this sub-family of orphan receptors (black box) (Ueda et al., 1992). The LBDs of Ftz-F1 proteins throughout the animal kingdom also share a high degree of similarity with several short blocks of identical amino acids scattered throughout the LBD. Of particular note is the highly conserved region at the C-terminus of Ftz-F1 proteins, which includes an AF-2 domain (Fig. 2A, reviewed in Glass and Rosenfeld, 2000). This suggests that the conservation within the DBD and LBD was responsible for the ability of β Ftz-F1 and *SF-1* to rescue $\alpha ftz-f1$ mutant phenotypes.

2.2. Evolutionarily conserved regions of Ftz-F1 are necessary for optimal interaction with Ftz protein

Ftz and Ftz-F1 bind cooperatively to DNA (Yu et al., 1997; Florence et al., 1997) and interact physically in Far Western assays (Guichet et al., 1997). In order to identify the regions of Ftz-F1 that directly interact with Ftz, the yeast two hybrid system was employed (Fields and Song, 1989; Chien et al., 1991). Sequential 5' and 3' deletions of $\alpha ftz-f1$ cDNA were fused to the GAL4 activation domain (AD) in pACT2. Full length *ftz* cDNA was fused to the GAL4 DBD in pAS2.1. Deletion from the 5' end of *ftz-f1* revealed three regions of the protein necessary for optimal interaction with Ftz (Fig. 3A). Deletion into the variable N-terminal A/B region (F1-393/1030) resulted in a precipitous drop in β -galactosidase activity, suggesting an interaction between the N-terminus of Ftz-F1 and full length Ftz. Further deletion of the A/B region had no effect on interaction with Ftz (F1-503/1030). However, truncations resulting in partial or complete removal of the DBD (F1-576/1030; F1-616/1030) resulted in a significantly decreased interaction. Thus, the DBD of Ftz-F1 appears to be necessary for strong interaction with Ftz protein. Finally, C-terminal fragments lacking the DBD still displayed considerable activity in the two hybrid assay (F1-616/1030; F1-946/1030), suggesting an additional interaction domain.

In keeping with this, deletion of the 73 amino acids at the C-terminus of Ftz-F1 in the 3' deletion series (Fig. 3B, F1-1/957) resulted in a dramatic decrease in interaction with Ftz. Further deletion from the C-terminus into the LBD had no additional effect (F1-1/702), localizing the region involved in Ftz interaction to the C-terminus of the Ftz-F1 LBD. Consistent with the 5' deletions, 3' deletion fragments that removed the entire DBD (F1-1/456) interacted weakly with Ftz, while a fragment retaining the N-terminal portion of the DBD (F1-1/517) displayed stronger interaction with Ftz. Finally, the N-terminal arm of Ftz-F1 was sufficient to interact weakly with full length Ftz protein (F1-1/456 and 1/265).

In sum, this deletion analysis identified three regions of Ftz-F1 that may directly contact Ftz: the N-terminal A/B region, the Ftz-F1 DBD and the C-terminus of the LBD which includes an AF-2 domain (see Fig. 2A). All three

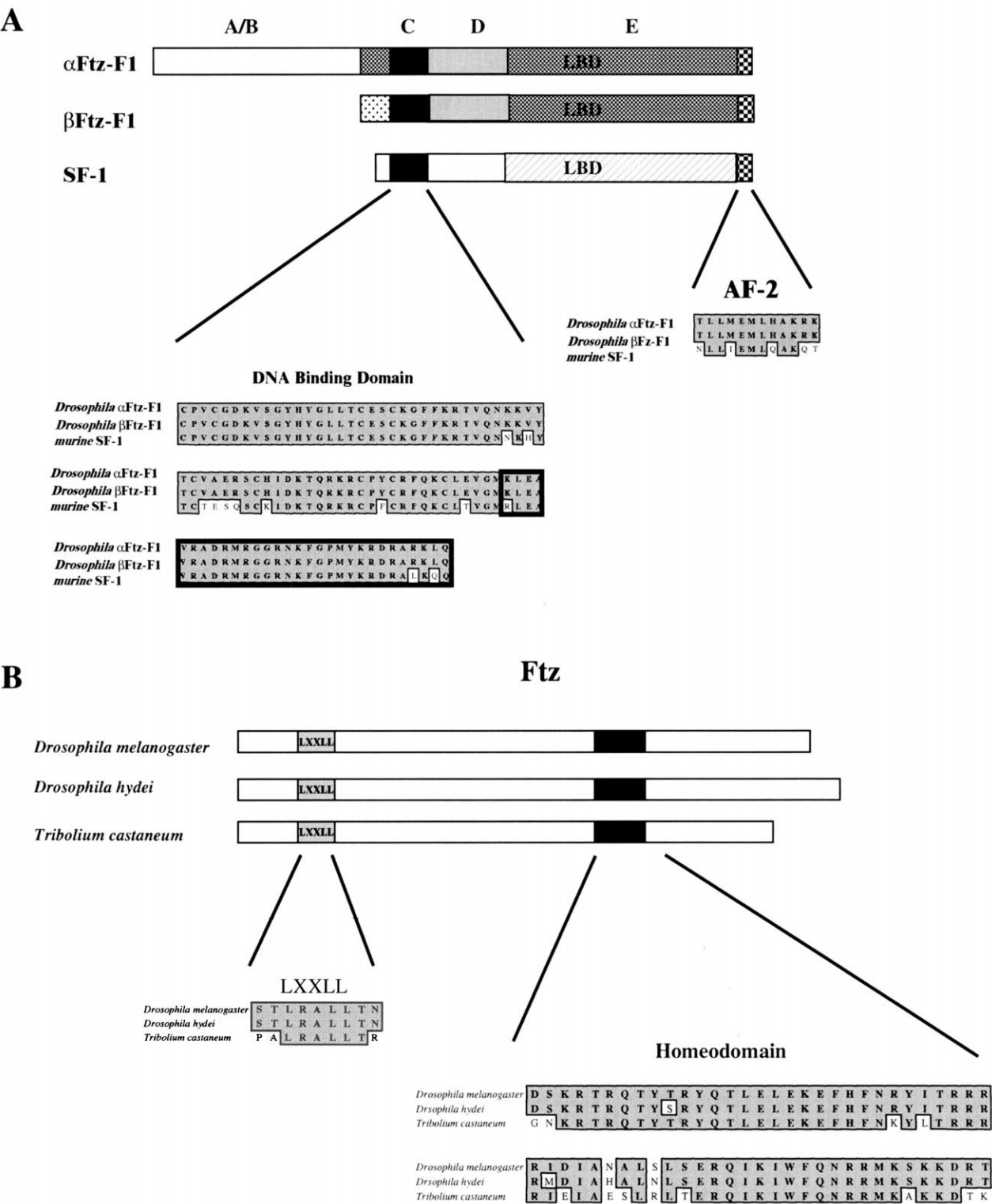


Fig. 2. Evolutionarily conserved domains of Ftz-F1 and Ftz. (A) Schematic diagram of the nuclear hormone receptor Ftz-F1. Schematic comparison of protein members of the Ftz-F1 family. The early and late forms of *Drosophila* Ftz-F1 are identical except for the N-terminal A/B region. The DBD of *Drosophila* Ftz-F1 proteins and murine SF-1 are highly conserved. The darker lined box within the DBD alignment indicates the Ftz-F1 box. Regions of similarity and identity are also dispersed through the LBDs of these proteins. Highlighted here is the conserved AF-2 domain. (B) Schematic comparison of Ftz protein from different species. Ftz from *D. melanogaster*, *D. hydei* and the beetle *T. castaneum* show a high degree of sequence identity within the HD. These proteins also share an identical hexapeptide (LRALLT) at the N-termini.

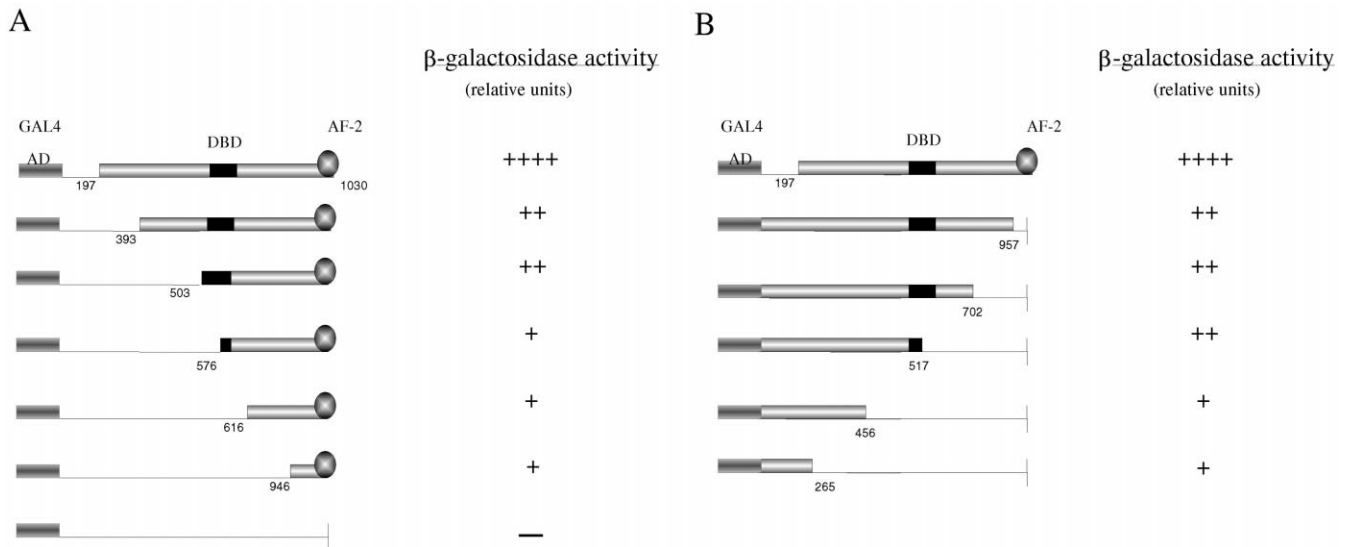


Fig. 3. Multiple regions of Ftz-F1 interact with Ftz. Sequential deletions of Ftz-F1 were generated and fragments were tested for interaction with full length Ftz fused to the GAL4 DBD in yeast two hybrid assays. (A) Schematic representation of 5' deletion constructs fused to the GAL4 AD and the relative β -galactosidase units generated by interaction with Ftz. (B) Schematic representation of 3' deletion constructs. The relative β -galactosidase units were grouped into the following ranges: (+++++) = >1500 units; (++++) = 500–1500 units; (+++) = 300–500 units; (++) = 150–300 units; (+) = 100–150 units; (–) $\leq$ 100 units. β -Galactosidase units were calculated as described in Section 4 based on measurement of at least six independent transformants. Note that neither full length pACT-Ftz-F1 nor pACT-Ftz-F1 deletions, activated reporter gene transcription alone, indicating that there are no Ftz-F1 binding sites in the promoter region.

regions are necessary to achieve optimal levels of protein–protein interaction, as deletion of any single domain weakened the Ftz-F1/Ftz interaction considerably. Two of these candidate Ftz-interaction domains are present in both early and late forms of *Drosophila* Ftz-F1, are present in Ftz-F1 family members throughout the animal kingdom (Fig. 2A) and are found in the Ftz-F1 orthologs that rescued α ftz-*fl* mutant embryos (Fig. 1). Thus, evolutionarily conserved regions of Ftz-F1 that are necessary for protein function in vivo act, at least in part, by mediating direct interactions with Ftz. This led us to further investigate the role of Ftz-F1's DBD and AF-2 portion of its LBD in the Ftz-F1/Ftz interaction, as described in the following two sections.

2.3. The Ftz-F1 DBD interacts directly with Ftz protein

The studies above demonstrated that the Ftz-F1 DBD was necessary for optimal interaction with Ftz protein. To test whether the Ftz-F1 DBD is sufficient to mediate an interaction with Ftz, yeast two hybrid assays were carried out. Surprisingly, the Ftz-F1 DBD interacted with Ftz almost as strongly as Ftz-F1 protein (Fig. 4A, slashed bars). These two hybrid results suggest that the Ftz-F1 DBD directly interacts with Ftz protein.

We showed previously that Ftz-F1 protein strongly enhances Ftz binding to an oligonucleotide containing adjacent Ftz/Ftz-F1 binding sites (Yu et al., 1997). Ftz binds DNA in the presence of Ftz-F1 at a concentration $\sim$ 50 times lower than that needed for Ftz to bind DNA alone. This results in the formation of a ternary Ftz/Ftz-F1/DNA

complex that is readily detectable in gel retardation assays. To test whether the Ftz-F1 DBD is sufficient to facilitate DNA binding by Ftz protein, purified Ftz-F1 DBD was included in DNA binding reactions with full length Ftz (Fig. 4B). Ftz-F1 DBD strongly enhanced Ftz binding to DNA, with formation of a ternary Ftz-F1 DBD/Ftz/DNA complex (lane 2, arrow) under conditions for which no binding of Ftz protein alone was detected (lane 5). The addition of anti-Ftz antibody, but not pre-immune serum, abolished the formation of this complex, demonstrating that Ftz protein is present in the ternary complex (lanes 3 vs. 4). Thus, the Ftz-F1 DBD is sufficient to dramatically enhance Ftz binding to DNA target sites.

We next tested whether interaction between the two DBDs – Ftz-F1 DBD and the Ftz HD was responsible for the facilitation of Ftz DNA binding seen above. Increasing amounts of Ftz HD were incubated with a fixed amount of Ftz-F1 DBD and DNA (Fig. 4C). The formation of a ternary complex was observed (Fig. 4C, arrow, lanes 5–7). Ftz-F1 DBD formed a binary complex with DNA (lanes 2–5) but all Ftz-F1 DBD was recruited to a ternary complex at high concentrations of Ftz HD (lane 7). In contrast to the situation for full length Ftz protein, ternary complex formation was detected only at levels of Ftz HD that also formed a HD/DNA binary complex (lanes 5–8). Thus, facilitation of binding was weaker with the isolated DBDs than with Ftz-F1 DBD and full length Ftz or with the two full length proteins. Nonetheless, some degree of synergy was observed since ternary complex formation would not be expected stochastically in the presence of excess DNA (lanes 5,6). This enhancement likely results from direct protein–protein

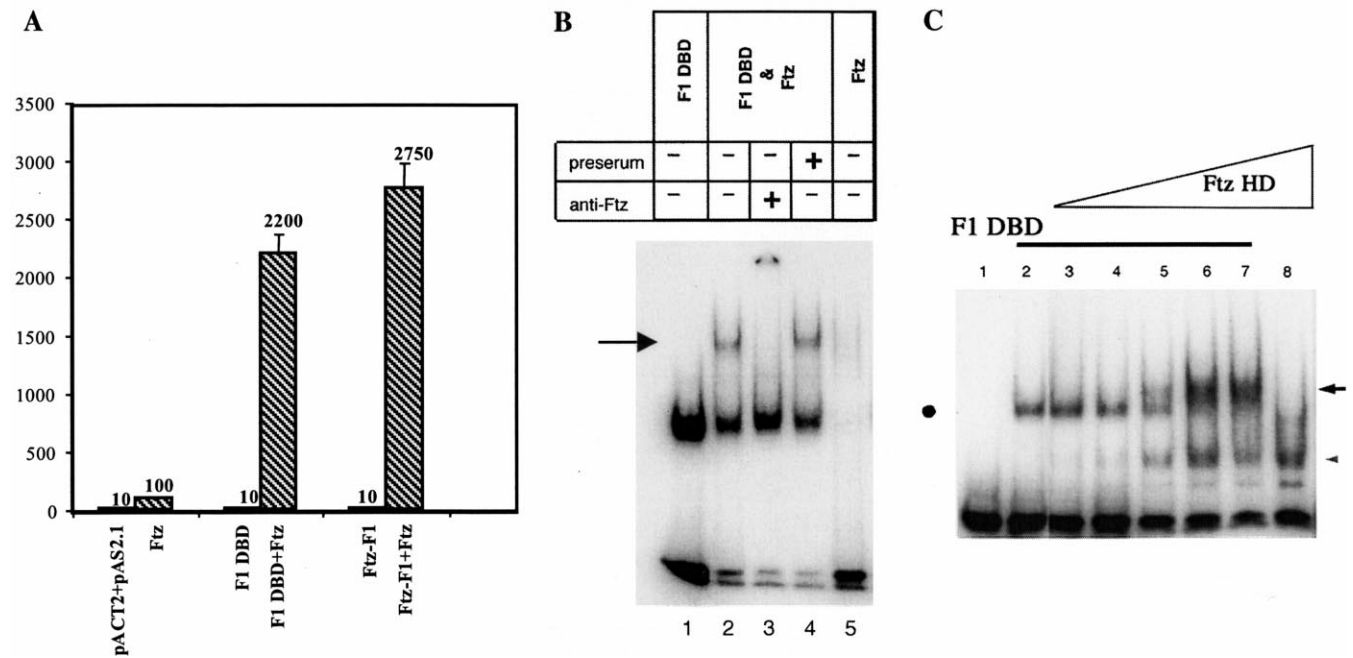


Fig. 4. The Ftz-F1 DBD is sufficient to mediate strong interactions with Ftz protein. (A) Interaction of Ftz-F1 DBD with Ftz in a yeast two hybrid assay. β -Galactosidase units were calculated based on measurement of six independent transformants. Columns indicate (from left to right): Controls, empty pAS2 + empty pACT; pAS-Ftz + empty pACT. Interaction of Ftz with Ftz-F1 DBD, empty pAS + pACT-F1 DBD; pAS-Ftz + pACT-F1 DBD. Interaction of Ftz with Ftz-F1, empty pAS + pACT-Ftz-F1; pAS-Ftz + pACT-Ftz-F1. pACT-Ftz-F1 and pACT-F1 DBD did not activate reporter gene transcription without Ftz. (B) F1 DBD facilitates Ftz binding to DNA. Gel retardation assays were used to compare the binding of Ftz to its target DNA (F1F) in the absence and presence of Ftz-F1 DBD. Lanes: (1) 0.25 μ g of Ftz-F1 DBD; (2–4) 0.25 μ g of Ftz-F1DBD; and 1 μ g of Ftz protein. Lane 2 shows the formation of a Ftz/F1DBD/DNA complex (arrow). The addition of anti-Ftz antibody abolished the complex (lane 3) while addition of preimmune serum had no effect (lane 4). Lane 5, Ftz alone (1 μ g) was not able to bind to F1F DNA. (C) F1 DBD and Ftz HD interact. Gel retardation assays were carried out with isolated F1DBD and Ftz HD. Increasing amounts of Ftz HD were incubated with a fixed amount of Ftz-F1 DBD. Lanes: (1) free probe, (2) 0.25 μ g of F1 DBD. Note formation of Ftz-F1/DNA binary complex (dot). Lanes 3–7, 0.25 μ g of F1 DBD and increasing amounts of Ftz HD (lane 3, 50 ng; lane 4, 100 ng; lane 5, 200 ng; lane 6, 450 ng; lane 7, 900 ng). In lane 5, note the formation of a ternary complex (arrow) and the appearance of binary Ftz HD/DNA binary complex (gray arrowhead). At high concentrations of Ftz HD, all F1 DBD is recruited to the ternary complex (lane 7). Lane 8, 900 ng Ftz HD.

contacts between the two DBDs. Alternatively, Ftz or Ftz-F1 may configure the DNA to increase binding by the partner protein. Such a mechanism contributes to cooperative DNA binding by the Hox–Extradenticle cofactor complex (Passner et al., 1999). In sum, these results indicate that the Ftz-F1 DBD interacts strongly and directly with Ftz protein. This interaction appears to involve weak contact between Ftz-F1 DBD and Ftz HD, but clearly requires additional region(s) of Ftz outside the HD that remain to be identified. The unexpected strength of interaction of Ftz-F1 DBD with full length Ftz, in both yeast two hybrid and gel retardation assays, suggests that the DBD of Ftz-F1 is sufficient to recruit Ftz to target DNA binding sites.

2.4. The Ftz-F1 AF-2 domain interacts with Ftz via an LXXLL motif

The Ftz-F1 deletion analysis (Fig. 3) suggested that, in addition to the DBD, the C-terminal portion of the Ftz-F1 LBD was necessary for maximal interaction with Ftz. This was further investigated in the yeast two hybrid system (Table 1). As mentioned above, the C-terminal portion of Ftz-F1 contains an AF-2 domain, expected to interact with LXXLL motifs in nuclear receptor coactivators. The Ftz N-

terminus contains a conserved LRALL sequence (Brown et al., 1994; Jost et al., 1995; Guichet et al., 1997; see Fig. 2B)

Table 1
Ftz-F1 interacts with the Ftz LXXLL motif and HD

Proteins added	β -Galactosidase units ^a
A. Ftz-F1 plus: Ftz	2000 ^b
Ftz N-terminus	235
Ftz LL AA	170
B. Ftz-F1-1/957 + Ftz N-terminus	<10
Ftz-F1 946/1030 + Ftz LL $\rightarrow$ AA	<10
C. Ftz-F1 plus: FtzdeltaHD	310
Ftz HD	215
Ftz LL AA/deltaHD	50
Empty pAS2.1	<10

^a β -Galactosidase units were calculated as described in Section 4. An average of results obtained from at least six independent transformants is shown. The Ftz proteins were expressed as fusions to the GAL4 DBD in pAS2.1. Units were adjusted to account for background levels of activation mediated by pAS2.1-Ftz. The Ftz-F1 proteins were expressed as fusions to the GAL4 AD in pACT2. These proteins did not activate transcription above basal levels in the absence of Ftz, as indicated.

^b This represents an underestimate of the Ftz/Ftz-F1 interaction as assays with the full length proteins reaches saturation.

and, as initially suggested by H. Krause, this region could mediate the interaction of Ftz with the Ftz-F1 AF-2 domain. As shown in Table 1A, an N-terminal fragment of Ftz, including the LRALL sequence, mediated a significant interaction with Ftz-F1 in the yeast two hybrid system. This N-terminal fragment displayed no detectable interaction with a Ftz-F1 protein lacking the C-terminal AF-2 domain (Table 1B), suggesting a direct interaction between the N-terminus of Ftz and the AF-2 domain of Ftz-F1. To directly test the function of the LRALL sequence, the LRALL sequence in full length Ftz was mutated to LRAAA. The interaction with Ftz-F1 was dramatically reduced by mutation of the LXXLL motif (Table 1A). The Ftz LXXAA failed to interact with a peptide containing only the Ftz-F1 C-terminus (F1-946/1030, Table 1B). These results indicate that there is a direct interaction between the AF-2 domain of Ftz-F1 and the LXXLL motif of Ftz.

While this manuscript was in preparation, Schwartz et al. (2001) reported that the Ftz LXXLL motif and the Ftz-F1 AF-2 domain can interact in Far-Western assays. These findings provide an additional in vitro method supporting the physical interaction between these motifs of Ftz and Ftz-F1. Schwartz et al. (2001) further showed that the LXXLL region was necessary for Ftz to generate *anti-ftz* phenotypes (Struhl, 1985) and that the AF-2 domain of Ftz-F1 is required for rescue of *ftz-f1* cuticle defects, further supporting the importance of this interaction for protein function in vivo.

2.5. Ftz HD interacts with Ftz-F1

Although the interaction was dramatically reduced, Ftz LXXAA still interacted with Ftz-F1, suggesting another interaction domain was still functional. To test the possibility that the HD was responsible for this interaction, a Ftz protein – Ftz delta HD – with a complete deletion of the HD³ was tested for interaction with Ftz-F1 (Table 1C). The interaction with Ftz-F1 was dramatically decreased by the deletion of the Ftz HD, suggesting that the HD contributes to interaction of Ftz with Ftz-F1. Note that mutation of the LXXLL motif or deletion of the HD each resulted in a roughly equivalent precipitous drop in interaction with Ftz-F1 in yeast two hybrid assays. The Ftz HD alone was sufficient to mediate a direct interaction with Ftz-F1 protein (Table 1C). As for the Ftz N-terminus alone, this interaction was significantly weaker than that of full length Ftz with Ftz-F1. To confirm that Ftz interacts with Ftz-F1 through two domains, we generated a double mutant Ftz carrying the LXXAA substitution and lacking the HD; interaction was reduced to near background levels, although residual interaction was still observed (Table 1C). These results indicate

that both the HD and the LXXLL motif of Ftz contribute to interactions with Ftz-F1 protein.

2.6. Ftz and Ftz-F1 coordinately activate transcription in vitro and in vivo

Ftz and Ftz-F1 are presumed to function in vivo by regulating the expression of target genes (reviewed in Biggin and McGinnis, 1997). We originally identified Ftz-F1 as a Ftz partner by virtue of their interaction on a bona fide Ftz target element from the *ftz* autoregulatory upstream element, the 323 bp *ftz* proximal enhancer (323 *fPE*) (Hiromi and Gehring, 1987; Pick et al., 1990; Schier and Gehring, 1992; Han et al., 1993; Yu et al., 1997). This autoregulatory element serves as a model Ftz/Ftz-F1 target element that functions in the same fashion as the Ftz-dependent enhancer of the downstream target *engrailed* (Florence et al., 1997). The 323 *fPE* contains five native binding sites for Ftz and three Ftz-F1 binding sites, including one adjacent Ftz/Ftz-F1 binding site to which the proteins bind cooperatively (Yu et al., 1997) (Fig. 4). To quantitate the ability of Ftz and Ftz-F1 to synergistically activate transcription via this native regulatory element, the 323 *fPE-lacZ* reporter was integrated into the yeast genome and full length Ftz and/or Ftz-F1 were expressed (Fig. 5A). Synergistic activation of transcription was seen when both proteins were expressed, leading to an ~50-fold activation of reporter gene expression. Ftz-F1 alone was able to activate transcription of the reporter, while Ftz alone was minimally active. The addition of Ftz increased levels of activation by Ftz-F1 alone 5–10 fold.

The 323 *fPE* directs reporter gene expression in a Ftz-dependent, *ftz*-like seven stripe pattern in vivo (Han et al., 1993 and Fig. 5B), reflecting the positive autoregulatory activity of this element. As shown in Fig. 5B, Ftz-F1 is also necessary for 323 *fPE-lacZ* reporter gene expression in vivo: in embryos derived from germline clones lacking *ftz-f1* activity, no expression of β -galactosidase was detectable, as is the case for *ftz* mutant embryos. This demonstrates that Ftz-F1 is absolutely required for activation of this Ftz-dependent target gene in vivo.

As mentioned above, the 323 *fPE* contains one composite Ftz-F1/Ftz site (F1F) to which Ftz-F1 and Ftz bind cooperatively (Fig. 4 and Yu et al., 1997). To test activation via this minimal binding site, Ftz-F1 and/or Ftz protein(s) were expressed along with a reporter gene containing three tandem copies of F1F upstream of a minimal promoter and *lacZ*. As for the native 323-*fPE* target element, Ftz and Ftz-F1 synergistically activated transcription. No activation above basal levels was detected without the F1F binding site (Fig. 5C). As for the native reporter, Ftz-F1 alone activated more strongly than Ftz alone and levels of Ftz-F1 activation were increased 5–10 fold by addition of Ftz. Interestingly, in keeping with the in vivo rescue results presented in Fig. 1, the murine ortholog SF-1 also interacted

³ FtzdeltaHD is a Ftz protein with a complete and exact deletion of the HD. A previously published Ftz protein, Ftz Δ HD, contains a deletion within the HD that removes helix 2 and parts of helices 1 and 3 (Copeland et al., 1996; Furukubo-Tokunaga et al., 1992).

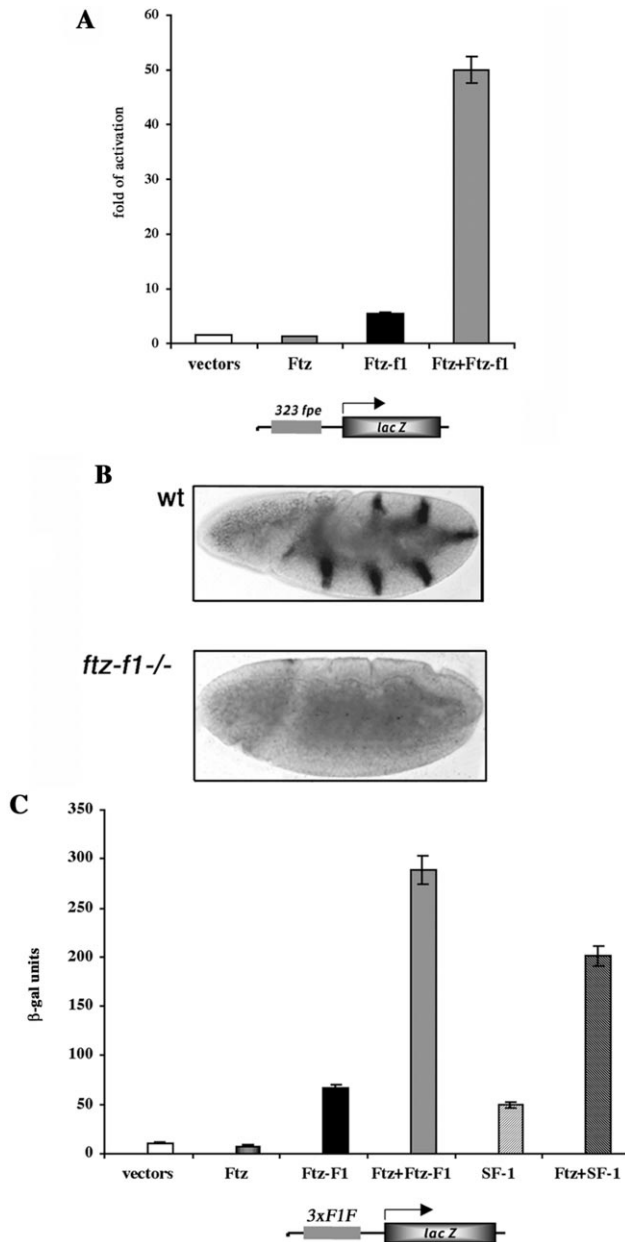


Fig. 5. Ftz and Ftz-F1 coordinately activate transcription in vivo and in vitro. (A) Transactivation of the 323 *fpe-lacZ* reporter gene in vitro. Yeast cells carrying an integrated copy of the 323 *fpe-lacZ* reporter were transformed with plasmids expressing Ftz and/or Ftz-F1, as indicated. β-Galactosidase activity was measured as described in Section 4 based upon assays performed with six individual transformants. (B) Expression of the 323 *fpe-lacZ* reporter gene in embryos requires Ftz-F1. The 323 *fpe-lacZ* fusion gene was expressed in seven *ftz*-like stripes in wild type embryos. In embryos derived from *ftz-f1*^{P1598} mutant germline clones, no reporter gene expression was detected. The expression was monitored with anti-β-galactosidase antibody (Cappell). The embryos are shown at the germ band extension stage; anterior, left. (C) The FIF binding site mediates transactivation by SF-1 and Ftz. The w3031A yeast cells carrying the reporter construct pSH201-3x*FIF-lacZ* were transformed with constructs expressing full length Ftz, Ftz-F1, Ftz + Ftz-F1, SF-1 or Ftz + SF-1, as indicated. No β-galactosidase activity was detected with a pSH201 vector lacking the FIF binding site for any combination of proteins. β-Galactosidase activity was measured as described in Section 4 based upon assays performed with six individual transformants.

synergistically with Ftz to activate transcription via the FIF composite site.

In sum, the 323-*fpe* autoregulatory element provides a model target element for studying the Ftz/Ftz-F1 interaction. In vivo, both Ftz and Ftz-F1 proteins are absolutely required for activation of target gene expression. In vitro, Ftz and Ftz-F1 cooperatively bind to DNA target sites and synergistically activate transcription. Murine SF-1 can substitute for Ftz-F1 activity in embryos, presumably because of its ability to synergistically activate target gene expression in conjunction with Ftz protein.

3. Discussion

Nuclear hormone receptors and Hox proteins form two large families of transcription factors that regulate a wide range of processes during development, differentiation and homeostasis of organisms throughout the animal kingdom. Here, we have presented evidence for a novel type of interaction between a nuclear receptor and a Hox protein in regulating segmentation during *Drosophila* embryogenesis. The orphan receptor Ftz-F1 and HD protein Ftz cooperate to promote the development of alternate body segments, presumably by activating expression of downstream target genes such as *engrailed*. Ftz and Ftz-F1 are each sequence specific DNA binding proteins with strong transcriptional ADs (Ftz: Fitzpatrick and Ingles, 1989; Han et al., 1989; Winslow et al., 1989; Ananthan et al., 1993; Colgan et al., 1993, 1995; Ftz-F1, for example: Lala et al., 1992; Tsukiyama et al., 1992; Ayer et al., 1993; Honda et al., 1993; Ellinger-Ziegelbauer et al., 1994; Ohno et al., 1994; Shapiro et al., 1996; Halvorson et al., 1996; Han et al., 1998; Nachtigal et al., 1998). However, neither is apparently able to function in *Drosophila* embryos to regulate target gene expression in the absence of its partner, since mutation of either protein results in lethality accompanied by identical pair-rule defects. Each protein is necessary but neither Ftz nor Ftz-F1 alone is sufficient to select and activate target gene transcription in the embryo. Why do Ftz and Ftz-F1 require partners in vivo to regulate gene expression? By analogy to the function of *βftz-f1* as a competence factor for the ecdysone response (Broadus et al., 1999), *αftz-f1* can be seen as a competence factor for a 'pulse' of Ftz expression in seven stripes in the blastoderm embryo (Fig. 6A).

Why does Ftz-F1 fail to activate FIF targets in the absence of Ftz protein? Ftz-F1 proteins are strong transcriptional activators in a variety of cell systems. In *Drosophila*, *ftz-f1* is maternally deposited and Ftz-F1 is found in all somatic nuclei before Ftz is expressed in seven stripes (Fig. 6A, green). Yet, Ftz-F1 does not detectably activate transcription in the absence of Ftz either temporally – before Ftz is expressed zygotically – or spatially – in regions of the embryo outside the seven Ftz stripes. Thus the presence of Ftz functions as an apparent 'on' versus 'off' switch to enable Ftz-F1 to activate transcription. We propose two

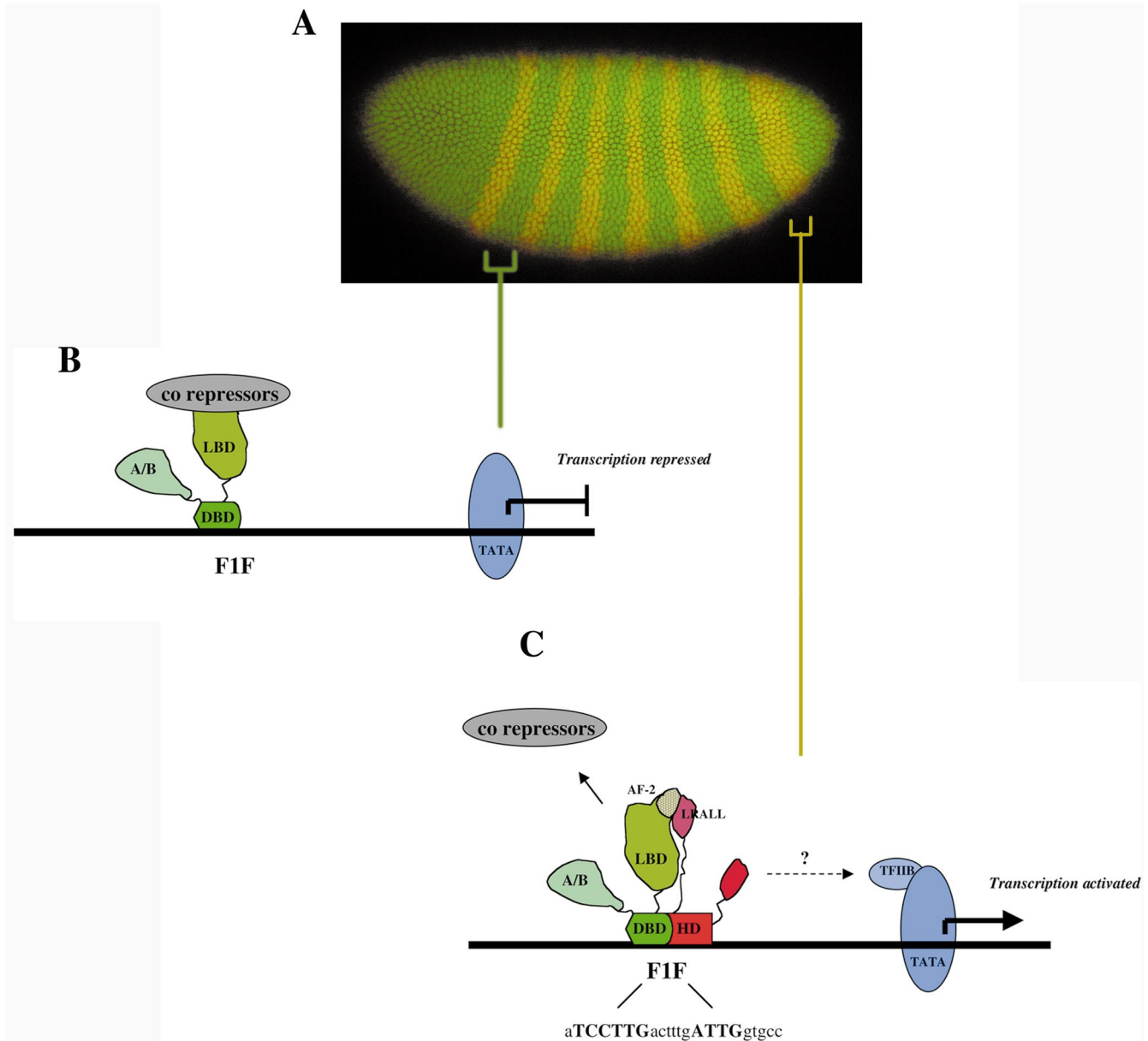


Fig. 6. Model for Ftz and Ftz-F1 interaction. (A) Ftz-F1 and Ftz overlap in the seven Ftz stripes. Ftz-F1 protein is expressed in all somatic cells of the blastoderm embryo (green). Ftz protein is expressed in seven stripes that encircle the embryo. Ftz protein was detected with a mouse anti-Ftz antibody (Kellerman et al., 1990) and rhodamine-conjugated secondary antibody. Ftz and Ftz-F1 overlap in all Ftz-expressing cells producing a yellow color. (B) Ftz-F1 is inactive in cells that do not express Ftz. Ftz-F1 protein is found in all somatic nuclei before Ftz is expressed in stripes and is found in the nuclei of all somatic cells when Ftz is expressed in a spatially restricted pattern of stripes. When Ftz-F1 is able to bind cognate DNA sites in the genome of these Ftz-non-expressing ('green') cells, we propose that corepressors are recruited to inhibit transcription. (C) Ftz-F1 and Ftz cooperate to activate target gene expression. Ftz and Ftz-F1 bind cooperatively to composite Ftz-F1/Ftz binding sites. Ftz-F1 strongly enhances Ftz binding and may thus recruit Ftz to sites where it was 'pre-bound' (B). Interaction of the Ftz LRALL sequence with the Ftz-F1 AF-2 domain displaces corepressors, allowing productive transcription complexes to form. Potential contacts of the Ftz C-terminus with TFIIB are indicated (Colgan et al., 1995). Ftz-F1 and Ftz interact synergistically through multiple protein domains. Ftz-F1 DBD interacts with Ftz protein very strongly and likely makes multiple contacts with Ftz; these sites remain to be identified and are not indicated in the schematic. The F1F binding site sequence indicated is derived from the *ftz* 323 proximal enhancer and functions in vivo to mediate Ftz-F1/Ftz dependent gene expression.

mechanisms that may contribute Ftz-F1's requirement for Ftz to activate transcription. First, Ftz-F1 might require Ftz for stable DNA binding. In vitro, Ftz-F1 activates transcription of the 323-*fPE* and F1F reporter constructs more strongly than does Ftz (Fig. 5), consistent with the in vitro DNA binding properties of the two proteins (Yu et al.,

1997). In fact, the order of magnitude of synergy of transcriptional activation of Ftz-F1 compared to Ftz-F1 + Ftz (5–10 fold, Fig. 1) is similar to the enhancement of DNA binding conferred by Ftz on Ftz-F1 (Yu et al., 1997). While this mechanism likely contributes to Ftz-F1 activation, the enhancement seen in vitro may not translate into the appar-

ent 'on' vs. 'off' state of Ftz-F1 in Ftz-expressing vs. Ftz-non-expressing cells in vivo. Therefore, we favor a model in which Ftz-F1 is actively repressed in the absence of Ftz, even if it is bound to cognate DNA target sites (Fig. 6B). Ftz-F1 is present in all somatic nuclei before Ftz is expressed and may bind cognate DNA sequences, but remains quiescent until it is activated by interaction with Ftz protein. We propose that interaction of Ftz with Ftz-F1 through its LXXLL motif displaces corepressor molecules, allowing productive transcription complexes to form. The candidates for repressor molecules that keep Ftz-F1 in an 'off' state are the corepressors that inhibit activity of other nuclear receptor family members. The corepressors identified in *Drosophila* that are expressed throughout the blastoderm embryo include Alien, which interacts with Ftz-F1 in GST-pulldown assays (Dressel et al., 1999) and SMRTER (Tsai et al., 1999). In Fig. 6, we have depicted a corepressor binding directly to Ftz-F1. It is also possible that another partner of Ftz-F1 recruits a corepressor to the complex, as has been shown for SF-1 which is regulated by interaction with the nuclear receptor Dax-1 that recruits corepressors and inhibits SF-1 activity (Crawford et al., 1998; Nachtigal et al., 1998).

Why does Ftz require Ftz-F1 for target gene regulation in vivo? As shown schematically in Fig. 6C, Ftz and Ftz-F1 bind cooperatively to DNA to select specific target sites that are composite binding sites for the two proteins. The binding of Ftz and Ftz-F1 to composite sites is stabilized by protein–protein interactions mediated by at least two regions of each partner protein. The composite nature of the binding site raises the selectivity of Ftz binding by requiring a heterodimeric site for productive interaction. The affinity of Ftz for composite sites is dramatically increased by Ftz-F1 (Yu et al., 1997; Florence et al., 1997 and Fig. 4). Ftz protein can bind to monomeric 'ATTA' core sites in vitro (Pick et al., 1990; Florence et al., 1991) and it appears to associate with a wide array of sites in the genome, as determined by UV cross-linking experiments (Walter et al., 1994; Biggin and McGinnis, 1997). However, we propose that in vivo, binding of Ftz to monomeric 'ATTA' sites is transient, as Ftz scans the genome for cognate DNA binding sites. This relatively unstable binding does not allow Ftz to effectively activate transcription on its own, consistent with the finding that concatamerized Ftz binding sites do not mediate a Ftz-dependent pattern of gene expression in vivo (Kalionis and O'Farrell, 1993). Ftz binds productively to composite heterodimeric sites, where it is stabilized by protein–protein interactions with Ftz-F1 (Fig. 6C). This notion is also consistent with reports that fusion of Ftz to a strong VP16 activator did not alter target gene regulation in vivo (Nasiadka et al., 2000). Thus, specificity of gene regulation by Ftz protein is achieved at the level of DNA binding/target site selection, as a result of interaction with DNA binding cofactors such as Ftz-F1.

Future studies are required to elucidate the detailed mechanism whereby transcription is activated by the Ftz/

Ftz-F1 complex. One candidate mechanism is direct contact of TFIIB by Ftz and/or Ftz-F1. Ftz was shown to directly contact TFIIB, activating transcription via a C-terminal region that does not appear to be involved in contacting Ftz-F1 (Colgan et al., 1993, 1995). Similarly, SF-1 interacts with TFIIB via the Ftz-F1 box and adjacent proline-rich region (Li et al., 1999), neither of which appear to be necessary for interaction with Ftz. Thus, these regions of the proteins could be available in the DNA bound ternary complex to directly contact the basal transcription machinery.

3.1. Does Ftz interaction substitute for a ligand?

We proposed above that Ftz interacts with the AF-2 domain of Ftz-F1 to mediate transcription activation. For other nuclear receptors, this activation step – unmasking of the LXXLL motif for coactivator binding – is a consequence of ligand binding. What is the situation for Ftz-F1? We propose four hypotheses: (1) The Ftz-F1 LBD adopts a unique structure such that it does not require a ligand-mediated conformational change to allow interaction with its AF-2 domain. However, deletion of part of the LBD enhances the transcription activity of SF-1, suggesting that the LBD does modulate SF-1 function (Shen et al., 1994). (2) As yet unidentified ligand(s) regulate Ftz-F1. Thus far, no candidate Ftz-F1 ligand has been identified in *Drosophila* and reports of a ligand for SF-1 appear to be controversial (Lala et al., 1997; Mellon and Bair, 1998). This does not exclude the possibility that an unknown cell-type restricted or ubiquitous ligand influences Ftz-F1. (3) The DNA mimics a ligand: binding of Ftz-F1 to DNA results in a conformational change that activates transcriptional potential. (4) Ftz mimics a ligand: multiple regions of Ftz-F1 interact with Ftz raising the possibility that one of these interactions mimics ligand binding in conferring a large conformational change in the LBD of Ftz-F1. A somewhat similar mechanism was proposed for interaction of SF-1 with the HD partner Ptx1 (Tremblay et al., 1999). Experiments are underway to determine which, if any, of these mechanisms regulate Ftz-F1 activity in vivo.

3.2. Does Ftz interaction obviate a requirement for a Ftz-F1 coactivator?

The data presented above suggest the novel possibility that Ftz substitutes for coactivator function for *Drosophila* Ftz-F1. For mammalian SF-1, standard coactivator interactions have been demonstrated in vitro and in cell culture, suggesting that SRC and CBP/p300 family proteins are partners of SF-1 (Crawford et al., 1997; Ito et al., 1998; Monte et al., 1998). *Drosophila* CBP has been well characterized (Akimura et al., 1997a,b; Waltzer and Bienz, 1999) and one p160/SRC-type coactivator, Taiman, was recently identified (Bai et al., 2000). Like Ftz, *Drosophila* CBP has one LXXLL motif while Tai has four such motifs, as is typical for mammalian coactivators. However, it is unlikely that

interaction with either of these coactivators is sufficient to activate Ftz-F1 since both dCBP and Tai are expressed ubiquitously in the embryo, including the cells where Ftz-F1 is apparently inactive (green cells, Fig. 6A). One interesting possibility is that dCBP acts as a corepressor in the context of Ftz-F1, as it has been shown to do for TCF (Waltzer and Bienz, 1998). Thus dCBP might silence Ftz-F1 by interacting with its AF-2 domain via an LXXLL motif. This interaction could be displaced by Ftz because of an intrinsically higher affinity of its LXXLL motif as compared to that of dCBP. Note that peptides with variations of the LXXLL motif have different affinities for AF-2 domains of nuclear receptors (Darimont et al., 1998; McInerney et al., 1998; Heery et al., 2001). Alternatively, Ftz may interact preferentially with Ftz-F1 because of the additional protein–protein and protein–DNA interactions that bring high levels of Ftz protein in close proximity of Ftz-F1, driving the interaction of its LXXLL motif with Ftz-F1.

3.3. Homeodomain nuclear receptor interactions

While the findings discussed above highlight the importance of the LXXLL domain for Ftz function, our results also demonstrated a requirement for the HD in the Ftz/Ftz-F1 interaction (Fig. 6). These results suggest that in addition to its role in DNA binding (Gehring et al., 1994), the Ftz HD is involved in direct protein–protein interactions with its cofactor Ftz-F1. These findings underscore the importance of the HD, which is absolutely required for the wild type function of Ftz (Schier and Gehring, 1992; Furukubo-Tokunaga et al., 1992; Schier and Gehring, 1993), as it is for other Hox proteins (Gehring et al., 1994; Gibson and Gehring, 1988; Gibson et al., 1990; Mann and Hogness, 1990; Zhao et al., 1996; McGinnis and Kuziora, 1994). Some years ago, a ‘HD-independent’ activity of Ftz was described (Copeland et al., 1996; Hyduk and Percival-Smith, 1996). These studies made use of a protein carrying a deletion within the HD (Δ HD) which removes helix 2 and portions of helices 1 and 3. It was shown that this protein is able to cause ‘anti-*ftz*’ phenotypes (Struhl, 1985) that result from mis-expression of Ftz with a heat inducible promoter (Copeland et al., 1996; Hyduk and Percival-Smith, 1996) and it was suggested that this protein can rescue *ftz*-dependent cuticle when similarly overexpressed (Copeland et al., 1996; Hyduk and Percival-Smith, 1996). In contrast to these results, Ftz Δ HD was unable to rescue *ftz* mutants when the protein was expressed under control of endogenous *ftz* regulatory elements (Furukubo-Tokunaga et al., 1992). In addition, even subtle mutations within this region of the HD abolished rescue potential (Furukubo-Tokunaga et al., 1992). Finally, Ftz Δ HD was unable to rescue any cuticular defects associated with *ftz* mutations when expressed using native *ftz* regulatory elements (Schier and Gehring, 1993). Thus it is likely that under conditions of overexpression, interactions with Ftz-F1 through the AF-2/

LXXLL domains of the proteins can to some extent overcome the endogenous requirement for the HD by positioning Ftz on the DNA of some target elements, allowing for gene activation.

Given the highly conserved nature of the HD, the finding that it is involved in the Ftz/Ftz-F1 interaction led us to ask if other Hox proteins can interact with Ftz-F1. Preliminary results indicate that a number of Hox proteins coordinately activate transcription in conjunction with Ftz-F1 in cells (unpublished observation). Experiments are underway to determine whether these interactions – which were less potent than Ftz – are strong enough to support interactions between Ftz-F1 and Hox proteins during *Drosophila* development. Such interactions would have been masked in previous genetic studies by the fact that the earliest function of Ftz-F1 in segmentation results in a phenotype which precludes analysis of Ftz-F1 function at slightly later stages when Hox genes such as Ubx, Antp and Scr are active. An additional question is whether conserved Hox proteins are partners of mammalian Ftz-F1 proteins. To our knowledge, co-expression of and interaction between Hox proteins and Ftz-F1 have not been investigated in mammals. Several partners of SF-1 have been characterized, including the nuclear receptor Dax-1, the zinc finger transcription factor WT-1 (Nachtigal et al., 1998), and the Bcd-family HD protein Ptx-1 (Tremblay et al., 1998). For Ptx-1, the interaction domain with SF-1 was mapped outside of the HD (Tremblay et al., 1999). However, interactions between other nuclear receptors and the HDs of other proteins have been reported. For example, the glucocorticoid, androgen and progesterone receptors interact with the POU domains of Oct-1 and Oct-2 (Prefontaine et al., 1998) and functional interactions between the glucocorticoid receptor and the HDs of Oct-1 and Oct-2 were demonstrated both in vitro and in vivo (Wang et al., 1999). Interestingly, the DBD of the glucocorticoid receptor interacts directly with several HDs, including the Ftz HD (Wang et al., 1999). Since *ftz-fl* but not *ftz* genes are found in the vertebrate lineage, an intriguing possibility consistent with the results discussed above is that vertebrate Ftz-F1 proteins interact with HDs of Hox proteins that are conserved throughout evolution, to execute unique functions as nuclear receptor/Hox protein heterodimers.

4. Experimental procedures

4.1. Plasmid construction

For transcription assays, cDNAs were cloned into pADNS (Colicelli et al., 1989) or Ycp50 ADH (Pick et al., 2000). The full length *ftz-fl* cDNA was reconstructed by joining a HindIII/ SalI fragment containing 57 bp of 5'UTR along with the first ~600 bp of the coding region, generated by PCR of genomic DNA, to a SalI/BamHI fragment from pJC20-*ftz-fl*. This fragment was inserted into

HindIII/NotI linearized pADNS followed by filling in of non-compatible BamHI-NotI ends before ligation. The *F1F-lacZ* reporter was generated by inserting an oligonucleotide containing three copies of F1F in the XbaI site of pSH201. The constructs to express proteins in bacteria were generated as follows: *ftz* cDNA was cloned into pET30a (Novagen) as an EcoRI fragment. The Ftz HD was generated by PCR and inserted into pET30. To express Ftz-F1, we used a previously described construct from C. Wu's laboratory, pJC20-Ftz-F1 (Lavorgna et al., 1991). The F1 DBD (aa510–605) was generated by PCR and inserted into pGEX. For yeast two hybrid assays, cDNAs were cloned into pAS2.1 or pACT2 (Clontech, CA). *ftz* cDNA was cloned into pAS2.1 as an EcoRI fragment. To generate the Ftz N-terminal fragment, *ftz* cDNA was deleted from the 3' side with the ExoIII/Mung Bean System (Stratagene). Site-specific mutagenesis was performed to generate Ftz deltaHD and the LRALL → LRAAA mutation with a Quickchange™ Kit (Stratagene) or with a modified protocol using Vent-Polymerase. The *ftz-f1* deletions were generated by PCR. The fragments were inserted into the BamHI/EcoRI sites of pACT2 (Clontech, CA) in frame with the Gal4 AD. For UAS constructs, *ftz-f1*, *βftz-f1* and *SF-I* cDNAs were each inserted in the EcoRI site of pUAS-T (Brand and Perrimon, 1993). All cloning steps used standard procedures; details are available upon request. All PCR products were verified by sequencing.

4.2. Gel retardation assays

Gel retardation assays were carried out as described by Han et al., (1993). Twenty microliter binding reactions containing 25 mM HEPES, pH 7.8, 0.5 mM DTT; 10% glycerol, ~10 fmol ³²P-labeled F1F oligonucleotide (5'-TCGAGCGGCATCCTTGACTTTGATTGGTGCCGC-3') and proteins as indicated, were incubated on ice for 1 h. The reactions were analyzed by electrophoresis through 4% native polyacrylamide gels. All proteins were synthesized in bacteria (strain BL21) and prepared as described (Yu et al., 1999). Ftz and Ftz HD were expressed with a His-tag and were purified using Niquel bound beads (Qiagen, CA). The F1 DBD was purified with a GST purification kit (Pharmacia, NJ). The anti-Ftz antibody was a rabbit polyclonal (Yu et al., 1997).

4.3. β-Galactosidase detection

Yeast strain SFY526 carrying the *lacZ* reporter gene (Clontech, CA) was used for the two-hybrid assays. The β-galactosidase assay protocol was modified from Steve Hanes. Individual colonies were inoculated in the proper selective liquid medium and incubated in the 30°C shaker incubator until saturation (OD ~ 1.0). The cultures were then diluted approximately 100× and incubated again at 30°C with agitation until log phase. The OD₆₀₀ of each culture was determined. The cells were collected and resuspended in 10 mM Tris/0.01% Triton X-100, immediately

frozen in dry ice and stored at –80°C. For assay, cells were thawed on ice and 1 ml of ONPG solution (0.01% *o*-nitrophenyl β-D-galactopyranoside, 2.5 mM DTT, 0.006% SDS, 40 mM β mercaptoethanol in Z-Buffer: 60 mM Na₂HPO₄, 40 mM NaH₂PO₄, 10 mM KCl, 1 mM MgSO₄) was added to each tube. The tubes were vortexed and incubated at 30°C. The color development was checked every 15 min. The reactions were stopped with addition of 0.3 ml of 1 M Na₂CO₃ solution. After brief centrifugation, OD₄₂₀ of each supernatant was measured. β-Galactosidase units were calculated as: OD₄₂₀/OD₆₀₀ × 1000/time (min). For transcription assays in yeast cells, w3031A cells were used. β-Galactosidase assays were carried out as described above. For the *323-fPE* reporter, incubation with ONPG was carried out overnight since levels of expression are lower than with the concatamerized reporter construct (Yu et al., 1997).

4.4. Transgenic experiments

P-element transformation was carried out using standard techniques. At least two independent lines were analyzed for each construct. The immunohistochemical detection and cuticle preparation used standard protocols. The embryos were photographed with a Zeiss Axiophot microscope using a digital camera (Diagnostic Instrument Inc., MI) with Nomarski or dark field optics.

Acknowledgements

We thank Keith Parker for the murine *SF-I* cDNA, Craig Woodard for the *βftz-f1* cDNA, Diane Duncan and Carl Wu for antibodies and Peter Gergen, Antoine Guichet and Anne Ephrussi for fly stocks. We are grateful to Steve Hanes for pSH201 plasmid and the β-galactosidase assay protocol. We thank Jianbo Song for generating the *ftz-f1* cDNA, Hong Zhu for technical assistance and Dave Kosman and Stephane Zaffran for advice. We thank Ron Kohanski, Craig Woodard, Reshma Taneja and Aneel Aggarwal for comments on the manuscript. This work was supported by a grant from the National Institutes of Health to L.P.

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